

CME Institute — Introduction to Futures

Source: CME Institute — "Introduction to Futures," Futures & Options Curriculum (CME Group, the listing exchange; primary source) — <https://www.cmegroup.com/education/courses>

What it is

A vendor-neutral, exchange-authored primer on how standardized, exchange-listed futures contracts actually work: how a contract is *specified*, how *margin* finances the position, how *volume and open interest* measure tradeability, and how *term structure and rolling* distort any continuous price series. The mechanics generalize cleanly to FX and crypto futures, so the framework compounds across instrument classes the screener may later add.

Core concepts

- **A future is a leveraged claim on notional, not a share.** One contract controls a large notional defined by a **multiplier**, so the quoted price alone is economically meaningless.
- **Standardization:** exchange fixes contract unit, tick size, tick value, listed months, and settlement, so contracts are fungible and centrally cleared.
- **Margin as performance bond:** you post a good-faith deposit (a small fraction of notional), and positions are **marked-to-market** daily; gains/losses flow as **variation margin**.
- **Two-dimensional liquidity:** each contract *month* has its own **volume** (flow) and **open interest** (outstanding stock).
- **Term structure & roll:** contracts across months trade at different prices (**contango/backwardation**), forcing a periodic **roll** that injects **roll yield** and price discontinuities into any stitched series.

Key metrics/signals it defines + how to read/compute them

- **Multiplier / notional:** notional = price × multiplier. E-mini S&P 500 (ES) multiplier = **\$50** (a 1-point move = \$50); Micro E-mini (MES) = **\$5**. WTI crude (CL) unit = **1,000 barrels**.
- **Tick size × tick value:** the minimum price increment and its dollar worth. ES tick = **0.25 index pts = \$12.50**; CL tick = **\$0.01/bbl = \$10.00**; Micro WTI (MCL) tick = **\$1.00**. Dollar-risk = (stop distance in ticks) × tick value.
- **Initial vs maintenance margin:** *initial* is required to open; *maintenance* is the floor — breach it and a **margin call** tops you back to initial. E.g., standard WTI maintenance ≈ **\$5,830**. Margin ÷ notional ≈ true leverage and capital-at-risk.
- **Volume + open interest:** volume = contracts traded/day (activity); OI = contracts still open (committed capital). Read together as the futures liquidity analog; near expiry, OI drains from the front month into the next — the roll signal.

- **Roll yield:** gain/loss from rolling. **Backwardation** → **positive** roll (roll down to cheaper far month); **contango** → **negative** roll (roll up).

How it maps to the Trader Screener

1. **§2g Suggested position size (ATR-based)** and **§2g Stop distance (ATR multiple)** must run through **dollar-risk-per-tick**, not share count: $\text{size} = (\text{account risk } \$) \div (\text{ATR-in-ticks} \times \text{tick value} \times \text{multiplier})$. This is exactly why a raw **§2a Last price** is uninterpretable for futures without the multiplier — 4,500 on ES is \$225,000 of notional, not \$4,500.
2. **§2c Average daily volume (ADV)** and **§2c Dollar volume (ADV × price)** should be redefined for futures as **contract volume + open interest of the active month**; OI is the tradeability filter that ADV alone misses, and it flags when the front month is dying (roll due). **§2c Bid/ask spread** and **§2c Market depth / book** remain [paid] real-time-only, unchanged.
3. **§2h IV term structure (contango/backwardation)** is the direct futures-curve analog — near vs. far month pricing — and it interacts with **§2a Return 1/3/6/12-month** and **§2a Relative strength rank**: a naïvely stitched (unadjusted) continuous series injects **phantom gaps** at each roll, faking momentum/breakouts. Back-adjusting removes the gap but corrupts absolute levels (and can go negative), so **§2a Moving averages / RSI / MACD** computed on the wrong series mislead. Also, **§2h Open interest** and **§2b Realized volatility** carry over as first-class futures fields.

Actionable takeaways

- Add a **multiplier** and **tick value** column per instrument; make every §2g sizing/stop calc consume them.
- For futures rows, replace ADV/dollar-volume with **active-month volume + OI**, and surface days-to-roll.
- Store a **continuous-series construction flag** (unadjusted / back-adjusted / return-spliced); default momentum/§2a metrics to a back-adjusted or return-spliced series and never to raw stitched prices.
- Compute **leverage = notional ÷ initial margin** and **capital-at-risk = maintenance margin** as display fields.

Open questions

- Which contract (standard vs. Micro) is the screener's default row per market?
- Where do per-contract spec/margin tables come from (CME feed vs. broker)?
- One canonical roll rule (OI-crossover vs. calendar vs. N-days-before-expiry) for all continuous series?
- Do we screen only the active month, or aggregate OI across the strip?

Sources

- CME Institute — Introduction to Futures course hub: <https://www.cmegroup.com/education/courses>
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